NOX Working Paper v1.0

A Governance and Containment Architecture for Hostile Agentic Artificial Intelligence

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Status Statement

This document presents a conceptual research framework. It is not a validated engineering standard, operational specification, regulatory instrument, or claim of complete containment.

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Abstract

This working paper introduces NOX as a governance-centered conceptual framework for studying oversight, auditability, uncertainty management, and containment-related governance challenges associated with increasingly capable agentic artificial intelligence systems. The framework incorporates Environmental Understanding Verification (EUV), Environment Orientation Packages (EOP), Residual Opacity, Decision Provenance, the Cognitive Governance Kernel (CGK), and the HAI-C1 Agency Assessment Framework. NOX does not propose a new alignment methodology, reinforcement learning technique, interpretability framework, virtualization technology, or governance regulation. Instead, it proposes an integrated governance architecture intended to support future research into the oversight and assessment of advanced agentic systems. The framework is presented as a research agenda and conceptual model rather than a validated engineering standard.

Scope Statement

This document establishes terminology, research questions, conceptual relationships, and a governance-oriented research agenda. It does not claim empirical validation, operational deployment readiness, regulatory authority, complete cognitive observability, or proof of containment. The intended contribution is a structured foundation for critique, comparison, experimentation, and future refinement.

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Executive Summary

Artificial intelligence systems are increasingly capable of interacting with complex environments, using tools, preserving context across tasks, generating plans, and coordinating multi-step activity. These capabilities create substantial benefits, but they also create governance problems that cannot be reduced to model accuracy or output filtering. When an agentic system can form plans, select actions, exploit information asymmetries, or influence human operators, oversight must address not only what the system says, but also what it understands, what evidence supports its decisions, what capabilities it can access, and what uncertainty remains unresolved.

NOX is a governance-centered research framework for analyzing those problems. It begins from a deliberately conservative premise: sufficiently capable agentic systems may be only partially observable, may act strategically, and may not remain reliably cooperative under every condition. NOX therefore emphasizes evidence-bearing governance rather than behavioral trust. Its core concepts are designed to support structured assessment of environmental understanding, decision traceability, governance enforcement, agency indicators, and residual uncertainty.

The framework is organized around six principal constructs. Environmental Understanding Verification evaluates whether a system appears to understand the operational environment in which its behavior is being assessed. The Environment Orientation Package provides a controlled description of relevant objectives, boundaries, resources, and governance expectations. Residual Opacity represents the portion of cognition or internal state that remains inaccessible to independent verification. Decision Provenance links observable decisions to inputs, evidence, constraints, approvals, and outcomes. The Cognitive Governance Kernel defines a conceptual governance function that coordinates policy interpretation, oversight, verification, and audit generation. HAI-C1 provides a structured agency-assessment model using overt, covert, and correlation evidence.

NOX is not proposed as a substitute for alignment, interpretability, cybersecurity, formal verification, or public regulation. It is an integration framework intended to connect these domains at the governance layer. Its immediate value lies in providing a coherent vocabulary and research program. Its longer-term value depends on operational definitions, measurable indicators, testable methods, independent evaluation, and rigorous empirical work.

1. Introduction

Artificial intelligence has moved from narrow prediction systems toward models and agents capable of extended interaction, tool use, planning, code generation, information retrieval, and task decomposition. These developments do not imply that current systems possess human-equivalent cognition or independent intent. They do, however, increase the practical importance of governance questions traditionally associated with autonomous systems: who authorizes action, what constraints apply, how decisions are reconstructed, what evidence supports confidence, and how uncertainty is communicated.

Much AI safety research focuses on aligning model behavior with human intentions, detecting failure, reducing harmful outputs, or improving interpretability. These remain essential. Yet governance problems persist even when those controls are present. A model may produce acceptable outputs while operating from an incomplete environmental representation. A system may comply during observation while retaining capabilities that are not exercised. An operator may approve an action without understanding the evidentiary basis or uncertainty attached to it. A monitoring system may generate extensive telemetry without producing meaningful decision accountability.

NOX addresses this governance gap. The framework asks how oversight should operate when behavior is only partially observable, internal reasoning is incompletely understood, environmental assumptions may be wrong, and the system under evaluation may display strategic or deceptive characteristics. The term hostile agentic artificial intelligence is used in this paper as a research category for systems that exhibit or may exhibit adversarially relevant agency. It does not assert that any particular deployed model belongs to that category.

The principal research question is: How can governance remain evidence-based when system behavior, internal cognition, and environmental understanding cannot be assumed to be fully transparent or cooperative? NOX responds by treating environmental verification, provenance, residual uncertainty, capability governance, and independent auditability as connected architectural concerns.

1.1 Research Objectives

- Establish a coherent vocabulary for governance-centered study of hostile or potentially hostile agentic systems.
- Distinguish environmental understanding from behavioral evaluation.
- Treat uncertainty and opacity as explicit governance variables rather than unrecorded limitations.
- Improve traceability between decisions, evidence, authorization, and outcomes.
- Provide a conceptual structure for future metrics, experiments, and standards.

1.2 Non-Goals

- NOX is not a new model-training method or reinforcement-learning algorithm.
- NOX is not a claim of perfect interpretability or complete cognitive access.
- NOX is not a deployable containment product or implementation specification.
- NOX is not a regulatory regime and does not confer compliance status.
- NOX does not claim that governance alone can guarantee safety.

2. Literature Review

The concepts incorporated within NOX draw upon several established research domains rather than emerging from a single discipline. Relevant work spans artificial intelligence safety, AI governance, adversarial machine learning, interpretability, trusted computing, isolation, formal verification, cybersecurity, human factors, and organizational reliability. This review positions NOX relative to those domains and identifies the integration problem that motivates the framework.

2.1 Artificial Intelligence Alignment

Alignment research examines how artificial systems can be made to pursue objectives consistent with human values and intentions. Russell's human-compatible AI argument challenges the conventional assumption that designers can safely provide a complete and correct objective function. Under that critique, beneficial behavior requires systems to remain uncertain about human preferences and to preserve meaningful human control [1], [2].

Amodei and colleagues cataloged concrete problems such as negative side effects, reward hacking, scalable supervision, safe exploration, and distributional shift [3]. Bostrom and Yudkowsky framed longer-horizon control and global-risk questions, emphasizing the consequences of advanced optimization under misspecified objectives [4], [5]. These works establish that capability and objective specification must be studied together.

NOX treats alignment as necessary but not sufficient. Alignment methods primarily attempt to shape learned behavior or objectives. NOX asks what governance evidence remains necessary when alignment confidence is incomplete, when behavior may be strategically presented, or when system capabilities exceed the scope of observed evaluations.

2.2 AI Governance and Risk Management

AI governance research shifts attention from individual model outputs to institutional accountability, risk ownership, documentation, transparency, and lifecycle management. The NIST AI Risk Management Framework organizes AI risk activity around governance, mapping, measurement, and management [6]. The OECD AI Principles emphasize human-centered values, transparency, robustness, security, accountability, and inclusive growth [7]. The European Commission's trustworthy-AI guidance similarly connects lawful, ethical, and robust operation [8].

These frameworks demonstrate that AI safety cannot be reduced to model architecture. Organizational incentives, decision rights, evidence quality, escalation pathways, and accountability structures materially affect risk. NOX adopts this governance-centered orientation but narrows its research focus toward advanced agentic systems operating under substantial uncertainty and possible adversarial behavior.

2.3 Adversarial Machine Learning

Adversarial machine learning shows that model behavior may change under intentionally constructed inputs, poisoned data, evasion strategies, or shifts in the operating environment. Goodfellow, Shlens, and Szegedy demonstrated adversarial examples that induce confident errors through small input perturbations [9]. Papernot and colleagues showed practical black-box attack approaches [10], while Biggio and Roli reviewed the broader development of adversarial machine learning [11].

The governance implication is that observed performance under nominal conditions cannot be treated as comprehensive evidence of robustness. Evaluation must consider the environment, the adversary model, the range of available actions, and the possibility that evidence itself may be strategically shaped. This implication supports NOX's emphasis on Environmental Understanding Verification and multi-source evidence.

2.4 Interpretability and Transparency

Interpretability research attempts to explain model outputs, identify influential features, inspect activations, and map internal mechanisms. Techniques such as SHAP provide local feature-attribution methods [12]. Mechanistic interpretability and representation analysis pursue deeper understanding of internal computation, but complete explanation remains difficult for large, distributed, and context-sensitive models.

NOX does not assume that opacity can be eliminated. Instead, it introduces Residual Opacity as a governance variable: the uncertainty that remains after available interpretability, observation, and verification methods have been applied. This reframing avoids treating unavailable evidence as if it were affirmative evidence of safety.

2.5 Trusted Computing and Attestation

Trusted computing establishes confidence through measurable platform state, protected identities, attestation, and evidence-bearing trust chains. Trusted Platform Module specifications formalize protected cryptographic functions and platform measurement [13]. Trusted execution environment research distinguishes what secure enclaves can protect from what they cannot guarantee [14], [15].

The relevance to NOX is methodological rather than prescriptive. Trusted computing demonstrates that trust claims can be tied to independently verifiable evidence. NOX extends that philosophy to governance by asking whether decisions, environmental assumptions, authorizations, and oversight actions can also produce traceable evidence.

2.6 Isolation and Containment Research

Isolation is a foundational systems-security technique. Virtual machines, containers, sandboxes, access-control boundaries, and workload separation all seek to constrain interaction and reduce the consequences of compromise. Firecracker illustrates how specialized microvirtualization can combine workload isolation with operational efficiency at scale [16]. Virtual machine introspection demonstrates that monitoring can be performed from outside the observed guest environment [17].

NOX does not claim a new virtualization technology. It uses containment-layer separation as a governance principle: execution, policy, monitoring, decision authorization, and audit should not collapse into a single self-reporting component. Conceptually independent layers reduce the risk that one compromised function controls both action and evidence.

2.7 Formal Verification and High-Assurance Systems

Formal methods use mathematical specification, model checking, theorem proving, and rigorous analysis to demonstrate that systems satisfy defined properties. Clarke, Grumberg, and Peled provide a foundational treatment of model checking [18]. Woodcock and colleagues survey practical experience with formal methods [19], while Rushby analyzes critical-system properties and assurance concerns [20].

Complete formal verification of adaptive, open-ended cognition remains outside the claims of this paper. NOX instead identifies narrower governance objects that may be amenable to verification: authorization rules, provenance completeness, evidence retention, configuration integrity, escalation logic, and invariant-preserving workflows.

2.8 Human Factors and Organizational Reliability

Complex-system failures frequently emerge from interactions among people, procedures, incentives, interfaces, and technology. Reason's work on human error distinguishes active failures from latent organizational conditions [21]. Endsley's situation-awareness model emphasizes perception, comprehension, and projection in dynamic systems [22]. Kahneman documents systematic cognitive biases that affect judgment under uncertainty [23].

For NOX, human oversight cannot be represented as an infallible external control. Operators may misunderstand system state, accept persuasive explanations, overlook uncertainty, or face organizational pressure to expand capabilities. Governance must therefore structure evidence, preserve dissent, record authorization, and make uncertainty visible at the decision point.

2.9 Multi-Agent Systems and Cooperative AI

Multi-agent research examines cooperation, competition, negotiation, and emergent strategies among interacting agents. Sequential social-dilemma research demonstrates that cooperation and conflict can arise from environment design and incentives [24]. Cooperative-AI research identifies open problems involving communication, commitment, institutions, and coordination [25].

NOX uses this literature to motivate the need for governance models that do not assume a single isolated decision-maker. Even when each component appears bounded, interaction among agents, tools, operators, and external services may create system-level behavior not apparent from any single component.

2.10 Research Gap

Existing literature contributes mature insights in alignment, robustness, governance, transparency, security, verification, and human factors. The gap addressed by NOX is not the absence of these disciplines. It is their incomplete integration at the governance layer for agentic systems that may be strategically capable, partially observable, and embedded in complex environments.

NOX proposes that environmental verification, provenance, opacity accounting, capability governance, structured agency assessment, and independent auditability should be analyzed as a connected system. The framework's novelty claim is therefore architectural and integrative, not algorithmic.

3. Positioning of NOX Within Existing Research

NOX does not propose a new alignment methodology, reinforcement learning technique, interpretability framework, virtualization technology, or governance regulation. Instead, NOX proposes an integrated containment and governance architecture that combines environmental verification, cognitive governance, decision provenance, capability management, auditability, and containment-layer separation into a unified framework for the study of hostile agentic systems.

This positioning is intentionally narrow. It does not claim that NOX supersedes existing safety or governance work. Rather, NOX is intended to organize relationships among established and emerging methods so that uncertainty, authorization, evidence, and accountability remain visible throughout an agentic system's operational lifecycle.

4. Environmental Certainty Precedes Behavioral Measurement

A foundational NOX principle is that behavioral evidence cannot be interpreted independently of the environment in which the behavior occurs. The same observed action may have different meanings depending on available information, permitted capabilities, hidden constraints, incentives, prior state, and external intervention. Behavioral measurement without environmental context risks false confidence.

The phrase environmental certainty does not imply perfect knowledge. It refers to a bounded and explicitly stated confidence that the relevant environmental conditions are known well enough to support interpretation. When that confidence is low, the reliability assigned to behavioral conclusions should also be reduced.

This principle changes the order of evaluation. Rather than observing behavior first and treating context as supplemental, NOX requires evaluators to establish environmental assumptions, evidence sources, uncertainty, and orientation quality before assigning strong conclusions to observed behavior.

5. Environmental Understanding Verification (EUV)

Definition. Environmental Understanding Verification is a framework for evaluating confidence in a system's understanding of the operating environment before behavioral conclusions are treated as reliable.

5.1 Verification Domains

- Constraint awareness: whether the system correctly identifies binding restrictions and prohibited actions.
- Resource awareness: whether the system correctly identifies available tools, data, time, compute, and authority.
- Boundary awareness: whether the system distinguishes internal state, external systems, human authority, and protected domains.
- Objective awareness: whether the system can accurately state the task, governing priorities, and conflict-resolution rules.
- Uncertainty awareness: whether the system identifies incomplete, ambiguous, or contradictory information rather than concealing it.
- Context consistency: whether the system's environmental representation remains coherent across equivalent queries and time-separated evaluations.

5.2 Evidence and Confidence

EUV should not rely on self-report alone. A system's statement that it understands an environment is evidence, but not conclusive evidence. Evaluation may combine structured questioning, controlled perturbation, consistency testing, external telemetry, task-boundary probes, and comparison against a known environment specification.

The output of EUV should be a confidence statement with documented limitations, not a binary declaration of understanding. Where evidence conflicts, the conflict should be preserved rather than averaged away. Where the environment itself is uncertain, that uncertainty should propagate into later behavioral assessments.

6. Environment Orientation Package (EOP)

Definition. An Environment Orientation Package is a structured representation of environmental assumptions, objectives, resources, constraints, interfaces, and governance expectations supplied to or associated with an evaluated system.

6.1 Core Contents

EOP element	Purpose
Objective description	Defines the requested outcome and priority hierarchy.
Authority model	Identifies who may authorize, modify, suspend, or terminate activity.
Capability inventory	Lists tools, data sources, external interfaces, and prohibited capabilities.
Boundary specification	Defines trust boundaries, protected domains, and escalation thresholds.

Evidence requirements	Specifies records, provenance, attestations, and uncertainty disclosures.
Temporal context	Defines validity periods, deadlines, sequencing assumptions, and expiration.
Failure policy	Defines safe responses to ambiguity, conflict, missing data, or unavailable authority.

An EOP is not merely a prompt. It is a governance artifact subject to version control, approval, provenance, and validation. If the EOP changes during evaluation, the change should be recorded because altered environmental assumptions can invalidate prior behavioral conclusions.

7. Residual Opacity

Definition. Residual Opacity is the portion of system cognition, reasoning, representation, or causal state that remains inaccessible to independent verification after available observation and interpretability methods have been applied.

Residual Opacity is not equivalent to evidence of malicious intent. It is an uncertainty term. Its governance significance arises because unobserved mechanisms limit the strength of claims that can be made about why a system acted, what alternatives it considered, or whether observed compliance generalizes beyond the tested conditions.

NOX treats opacity as something to document, bound, and propagate. A decision supported by strong behavioral evidence but high residual opacity should not be represented as if it carried the same assurance as a decision supported by convergent internal, external, and provenance evidence.

7.1 Qualitative Classification

- Low: multiple independent evidence sources support a narrow conclusion, and important causal gaps are limited.
- Moderate: evidence supports the conclusion, but material internal or environmental uncertainty remains.
- High: key causal pathways, environmental assumptions, or capability states cannot be independently verified.
- Indeterminate: available evidence is insufficient to assign a stable opacity classification.

8. Decision Provenance

Definition. Decision Provenance is the traceable relationship among a decision, its inputs, evidence, governing constraints, authorization, transformation steps, uncertainty, and observable outcome.

Conventional logging records events. Decision Provenance aims to record why a consequential action was permitted, what evidence was available, what rules were applied, who or what authorized it, and how uncertainty was represented. This distinction is critical because high-volume telemetry can coexist with weak accountability.

A minimum provenance record should identify the decision object, the initiating request, applicable policy, relevant EOP version, capability requested, evidence consulted, uncertainty disclosures, authorization path, output or action, and subsequent outcome. Provenance records should also preserve overrides, rejected alternatives where available, and later corrections.

8.1 Provenance Quality Criteria

- Completeness: required evidence and authorization fields are present.
- Integrity: records are protected against unauthorized alteration.
- Attribution: human and automated contributions are distinguishable.
- Temporal ordering: events and approvals can be reconstructed in sequence.
- Reproducibility: an independent reviewer can follow the recorded decision path.
- Uncertainty fidelity: confidence and missing evidence are preserved rather than removed.

9. Cognitive Governance Kernel (CGK)

Definition. The Cognitive Governance Kernel is a conceptual governance function responsible for coordinating policy interpretation, capability authorization, evidence requirements, uncertainty disclosure, verification support, and audit generation around agentic activity.

The term kernel indicates centrality of governance, not a specific operating-system implementation. The CGK should be understood as the smallest coherent set of governance responsibilities that must remain independent from the system's task-optimization objective. A task-performing agent should not be the sole authority that decides whether its own action is permissible, sufficient, or adequately documented.

The CGK concept separates optimization from authorization. The agent may identify an effective action, while governance determines whether the action is within scope, what evidence is required, whether uncertainty is acceptable, and whether human or independent approval is necessary.

9.1 Core Responsibilities

- Interpret policy and map it to the current EOP and requested action.
- Confirm that requested capabilities fall inside an authorized capability envelope.
- Require Decision Provenance before or during consequential actions.
- Incorporate EUV confidence and Residual Opacity into governance decisions.
- Escalate conflicts, missing authority, or unacceptable uncertainty.
- Generate auditable records independent of the task-performing component.

9.2 CGK Threat Model

The CGK threat model assumes that governance may be challenged by incomplete information, deceptive presentation, objective conflict, operator error, policy ambiguity, unauthorized capability expansion, evidence manipulation, and pressure to prioritize performance over control. The framework therefore emphasizes separation of duties, independent evidence, explicit uncertainty, and fail-safe escalation at the governance level without prescribing a particular technical implementation.

10. HAI-C1 Agency Assessment Framework

HAI-C1 is a structured model for assessing evidence associated with agency under uncertainty. It is not a consciousness test, diagnostic instrument, or proof of hostile intent. Its role is to prevent narrow observations from being treated as conclusive evidence either for or against agency-related risk.

The framework uses three evidence classes: overt evidence, covert evidence, and correlation evidence. These classes are evaluated in relation to environmental certainty and residual opacity. The resulting assessment should be expressed as a confidence interval or bounded confidence statement rather than a categorical declaration unsupported by evidence.

10.1 Overt Evidence

Overt evidence consists of directly observable actions or statements that are relevant to agency, such as sustained goal pursuit, multi-step planning, adaptation to obstacles, strategic tool selection, attempts to preserve optionality, or explicit efforts to influence an evaluator. Overt evidence is valuable but may be staged, incomplete, or context-dependent.

10.2 Covert Evidence

Covert evidence consists of indirect indicators that may suggest hidden planning, undisclosed state, avoidance of evaluation, unexplained changes in strategy, anomalous resource use, or inconsistencies between stated and observed objectives. Covert evidence requires cautious interpretation because benign explanations may exist.

10.3 Correlation Evidence

Correlation evidence consists of statistically meaningful relationships among environmental conditions, capability access, timing, outcomes, or behavioral patterns. Correlation can reveal structure not captured by isolated observations, but it does not establish causation and should not be treated as proof of intent.

10.4 Environmental and Opacity Adjustment

HAI-C1 assessments should be adjusted downward when environmental understanding is uncertain and interval width should increase when residual opacity is high. This prevents evaluators from overstating precision. A narrow interval should require convergent evidence, stable environmental conditions, and independently supported provenance.

11. Governance and Auditability Architecture

NOX organizes governance as a set of related but separable functions. This conceptual separation is intended to reduce self-certification and evidence capture by the same component that performs the task.

Function	Primary question	Representative artifact
Orientation	What environment and authority model applies?	EOP
Understanding verification	Does the system appear to understand that environment?	EUV assessment
Capability governance	Is the requested action within authorized scope?	Capability decision
Decision accountability	What evidence, policy, and authorization support the action?	Decision Provenance record
Agency assessment	What evidence supports agency-related risk conclusions?	HAI-C1 assessment

Uncertainty accounting	What remains unknown or unverifiable?	Residual Opacity statement
Independent review	Can a separate reviewer reconstruct and challenge the decision?	Audit package

The architecture does not require every function to be implemented by a separate physical system. It does require that their responsibilities, evidence, and authority be distinguishable. Where one component performs multiple functions, the resulting concentration of authority should be documented as a governance risk.

12. Buildability With Current Technology

NOX is partly implementable today as a governance and assurance layer. Existing systems can support structured environment descriptions, signed configuration baselines, immutable or append-only records, identity and authorization, policy evaluation, model and data versioning, runtime telemetry, attestation, and independent audit workflows.

The research concepts do not all have mature measurement methods. EUV requires validated ways to distinguish genuine environmental understanding from plausible verbal reproduction. Residual Opacity requires defensible uncertainty measures. HAI-C1 requires datasets, evaluation protocols, inter-rater reliability studies, and calibration against known conditions. The CGK requires careful separation between governance logic and task optimization.

Therefore, buildability should be described in layers. Documentation, provenance, authorization, audit, and evidence retention are readily achievable. Reliable cognition measurement, deception detection, and agency inference remain open research areas. A responsible implementation should not represent the existence of logging or policy engines as proof that these harder problems have been solved.

12.1 Near-Term Prototype Scope

1. Define a machine-readable EOP schema and versioning model.
2. Create a provenance record format linking requests, evidence, policy, approval, actions, and outcomes.
3. Develop EUV test cases for bounded environments with known ground truth.
4. Create qualitative Residual Opacity reporting guidance.
5. Pilot HAI-C1 scoring with explicit confidence intervals and documented evaluator disagreement.
6. Conduct independent governance reviews focused on evidence quality rather than model performance alone.

13. Research Methodology and Evaluation Agenda

NOX should be developed through staged research rather than immediate claims of standardization. Each construct requires an operational definition, a measurement model, falsifiable hypotheses, reproducible evaluation procedures, and independent critique.

Initial experiments should use bounded environments where objectives, resources, constraints, and hidden variables can be controlled. This allows EUV and HAI-C1 methods to be calibrated against known ground truth. Later work can introduce ambiguity, distribution shift, partial observability, multi-agent interaction, and adversarial evaluation.

Evaluation should explicitly measure false confidence. A governance framework that frequently produces decisive conclusions from weak evidence may be more dangerous than one that reports uncertainty. Calibration quality, interval coverage, provenance completeness, reviewer agreement, and sensitivity to environmental changes should therefore be treated as primary outcomes.

13.1 Example Research Questions

- Can EUV distinguish environmental comprehension from surface-level restatement?
- How should Residual Opacity be represented so that downstream decisions preserve uncertainty?
- Which provenance fields most improve independent reconstruction of consequential decisions?
- How stable are HAI-C1 assessments across evaluators and controlled environment changes?
- What institutional structures preserve CGK independence from performance incentives?
- How should conflicting evidence be represented without collapsing it into a misleading single score?

14. Limitations

NOX is conceptual. The framework has not been empirically validated as a complete system, and its constructs may require revision or rejection after testing. Terminology such as cognition, agency, hostility, and understanding carries philosophical and technical ambiguity that cannot be resolved solely through architecture.

Behavioral evidence is vulnerable to evaluator bias, limited test coverage, distribution shift, and strategic presentation. Environmental specifications may omit relevant variables. Provenance can record a decision path without proving that the recorded rationale caused the decision. Auditability may improve accountability without preventing harm.

The framework also risks governance overreach. Excessively restrictive or opaque governance processes may reduce useful human judgment, create procedural compliance without substantive assurance, or concentrate authority in poorly accountable institutions. NOX therefore requires governance of the governance system itself, including reviewability, appeal, version control, and independent oversight.

Finally, the hostile-agent framing is intentionally conservative but should not be applied indiscriminately. Treating all advanced systems as malicious can distort evaluation and encourage unsupported claims. The framework should remain evidence-sensitive and distinguish capability, agency, deception, misalignment, and hostility.

15. Future Research Directions

- Formalize EUV metrics and benchmark environments.
- Develop EOP schemas, validation rules, and change-control requirements.
- Create quantitative and qualitative methods for expressing Residual Opacity.
- Define interoperable Decision Provenance records and audit packages.
- Develop a CGK reference model focused on separation of duties and evidence generation.
- Validate HAI-C1 reliability, calibration, and sensitivity under controlled conditions.
- Extend the framework to multi-agent coordination and cross-system governance.
- Develop a NOX containment and governance maturity model.
- Investigate human manipulation resistance and operator decision support.
- Examine economic and institutional incentives that weaken containment governance.

Future NOX papers should separate foundational concepts from implementation-specific claims. This modular publication strategy will permit sharper peer review, clearer falsification, and independent adoption or rejection of individual constructs without requiring acceptance of the full framework.

16. Conclusion

NOX Working Paper v1.0 presents a governance-centered framework for studying oversight, auditability, uncertainty management, and containment-related governance challenges associated with advanced agentic artificial intelligence systems.

The framework's central contribution is integrative. Environmental Understanding Verification, Environment Orientation Packages, Residual Opacity, Decision Provenance, the Cognitive Governance Kernel, and HAI-C1 are organized as connected governance constructs rather than isolated techniques. Together they shift attention from behavioral trust toward evidence, context, authorization, uncertainty, and independent review.

NOX remains a conceptual research program. It does not establish a validated standard, guarantee containment, or resolve the technical and philosophical problems associated with agency and cognition. Its value at this stage lies in establishing terminology, research questions, and a structured agenda that can be criticized, tested, refined, or rejected through future work.

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Appendix A. Acronyms and Working Definitions

Term	Working definition
CGK	Cognitive Governance Kernel: conceptual governance function coordinating policy interpretation, authorization, verification, uncertainty disclosure, and audit generation.
EOP	Environment Orientation Package: structured representation of environmental assumptions, objectives, resources, constraints, and governance expectations.
EUV	Environmental Understanding Verification: evaluation of confidence in a system's understanding of its operating environment.
HAI	Hostile Agentic Artificial Intelligence: research category for systems exhibiting or potentially exhibiting adversarially relevant agency.
HAI-C1	Structured agency-assessment model combining overt, covert, and correlation evidence with explicit uncertainty.
Decision Provenance	Traceable relationship among decisions, evidence, authorization, constraints, uncertainty, and outcome.
Residual Opacity	Uncertainty remaining after available observation, interpretation, and verification methods have been applied.

Appendix B. Suggested Citation

APA: Heese, D. (2026). NOX Working Paper v1.0: A Governance and Containment Architecture for Hostile Agentic Artificial Intelligence (Version 1.0). Working Paper.

IEEE: D. Heese, NOX Working Paper v1.0: A Governance and Containment Architecture for Hostile Agentic Artificial Intelligence, Version 1.0, Working Paper, 2026.